

Name: ....., NEPTUN-code .....

Group, teacher: .....

*Computer Science BSc Basic Mathematics TEST-1*  
*17-th of October, 2025*

*Reasoning and justification are needed in the solutions.*

1. (7 points) Find the simplest form of the following expression:

$$\left(\frac{b^2}{a^3 - ab^2} + \frac{1}{a + b}\right) : \left(\frac{a - b}{a^2 + ab} - \frac{a}{b^2 + ab}\right)$$

2. (8 points) Solve the following equation on the set of real numbers:

$$\sqrt{1 - x} - \sqrt{x + 19} = 2$$

3. (9 points) Solve the following inequality on the set of real numbers (here  $\lg$  denotes  $\log_{10}$ ):

$$\frac{1}{1 + \lg(-x)} + \frac{3}{3 + \lg(-x)} \leq 2$$

4. (4 points) Determine the parameter  $p$  so that  $x_0 = 3$  will be the root of the polynomial

$$P(x) := 2x^3 + px^2 + 2x + 3$$

Factor out the root factor to  $x_0$ . Determine the multiplicity of the root  $x_0$ .

5. (7 points) Solve the following equation on the set of the real numbers:

$$\tan x + \cot x = 2 \cdot \sin(2x)$$

6. (1+6+1=8 points) a) Give the mathematical form (using quantifiers) of the following statement:

- For all large enough natural numbers  $n$  we have:

$$\frac{10n^8 - 3n^5 + 2n^4 - 1990n^3 + n^2 + 111n - 7}{3n^5 - 31n^4 + 2n^3 - 7n^2 + 3n + 22} > 2025$$

- b) Prove that the statement is true.

- c) Give its negation too.

7. (7 points) Using mathematical induction prove the following statement:

$$\forall n \in \mathbb{N}^+ : \sum_{k=1}^n \frac{k \cdot 2^k}{(k+2)!} > \frac{1}{2} - \frac{2^{n+1}}{(n+2)!}$$